

1. What is Python?

- Python is a high-level, interpreted, object-oriented, and general-purpose programming language.

2. Keywords

- Keywords are **reserved words** in Python that have a predefined meaning. We cannot use them as variable names, function names, or class names.
- Python has exactly 35 reserved .
- **Constants:** True, False, None
- **Operators:** and, or, not, is, in
- **Control Flow:** if, elif, else, for, while, break, continue, pass
- **Functions & Classes:** def, return, lambda, yield, class
- **Exception Handling:** try, except, finally, raise, assert
- **Imports & Namespaces:** import, from, as, global, nonlocal, del
- **Context Management:** with
- **Async Programming:** async, await

3. Identifiers

- Identifiers are the names given by programmers to variables, functions, classes, or objects.
- Ex

```
student_name = "mohit"

age = 20

def calculate_salary():
    print("salary")
```

4. Variables

- A variable in Python is simply a name that stores a value in memory.
- Ex

```
num = 10      # num is a variable storing the number 10
name = "mohit" # name is a variable storing text
```

- Variable Naming Rules
 - 1) Must start with Letter or Underscore.
 - 2) Can contain Letters, Numbers, Underscore.
 - 3) No spaces.
 - 4) No special symbols.
 - 5) Cannot use keywords
- Good Naming Example

```
student_name = "mohit"

employee_salary = 4000

total_marks = 78
```

5. Comments

- A comment in Python is a piece of text in your code that is ignored by the Python interpreter. It's written only for humans to read, Programmers use comments to explain what the code does.
- Ex

```
# This is a single-line comment
x = 10 # This sets x to 10
```

```
"""
This is a multi-line comment.
Useful for longer explanations.
"""
```

6. Indentation

- **Indentation in Python** means adding spaces at the beginning of a line to show which block of code it belongs to.
- Python uses indentation instead of {} braces.

Other language	Python
if(age>18){ console.log("Eligible") }	if age > 18: print("Eligible")